

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8"/>
5   <meta name="description" content="Data Visualisation"/>
6   <meta name="keywords" content="HTML, CSS, D3"/>
7   <meta name="author" content="Anh Khoa"/>
8   <title>LAB 5.3 D3 Adding and Removing Data</title>
9   <script src="https://d3js.org/d3.v7.min.js"></script>
10 </head>
11
12 <body>
13   <h1>The D3 Journey starts here...</h1>
14
15   <!-- Buttons for different transitions -->
16   <div style="margin-bottom: 20px;">
17     <button id="addData">Add</button>
18     <button id="removeData">Remove</button>
19   </div>
20
21   <script>
22     var w = 500;
23     var h = 200;
24     var maxValue = 25;
25
26     var dataset = [14, 5, 26, 23, 9, 12, 18, 25];
27
28     var svg = d3.select("body")
29       .append("svg")
30       .attr("width", w)
31       .attr("height", h);
32
33     const xScale = d3.scaleBand()
34       .rangeRound([0, w])
35       .paddingInner(0.1);
36
37     const yScale = d3.scaleLinear()
38       .domain([0, maxValue])
39       .range([h, 0]);
40
41     function updateChart() {
42       xScale.domain(d3.range(dataset.length));
43
44       // === BARS ===
45       const bars = svg.selectAll("rect")
46         .data(dataset, (d, i) => i);
47
48       bars.exit()
49         .transition()
50         .duration(500)
51         .attr("x", w)
52         .remove();
53
54       const newBars = bars.enter()
55         .append("rect")
56         .attr("x", w) // xuất hiện từ ngoài khung
57         .attr("y", d => yScale(d))
58         .attr("width", xScale.bandwidth())
59         .attr("height", d => h - yScale(d))
60         .attr("fill", "orange");
61
62       newBars.merge(bars)
63         .transition()
64         .duration(500)
65         .attr("x", (d, i) => xScale(i))
66         .attr("y", d => yScale(d))
67         .attr("width", xScale.bandwidth())
68         .attr("height", d => h - yScale(d));
69
70       // === LABELS ===
71       const labels = svg.selectAll("text")
72         .data(dataset, (d, i) => i);
73
74       labels.exit()
75         .transition()
76         .duration(500)
77         .attr("x", w)
78         .remove();
79
80       const newLabels = labels.enter()
81         .append("text")
82         .text(d => d)
83         .attr("x", w)
84         .attr("y", d => Math.max(yScale(d) - 5, 10))
85         .attr("text-anchor", "middle")
86         .attr("font-size", "12px")
87         .attr("fill", "black");
88
89       newLabels.merge(labels)
90         .transition()
91         .duration(500)
92         .text(d => d)
93         .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2)
94         .attr("y", d => Math.max(yScale(d) - 5, 10));
95     }
96
97     // Initial draw
98     updateChart();
99
100    // Event handlers
101    d3.select("#addData").on("click", function () {
102      const newValue = Math.floor(Math.random() * maxValue);
103      dataset.push(newValue);
104      updateChart();
105    });
106
107    d3.select("#removeData").on("click", function () {
108      if (dataset.length > 0) {
109        dataset.shift();
110        updateChart();
111      }
112    });
113  </script>
114
115  <footer style="color: grey; margin-top: 20px;">
116    COS3045 Data Visualisation<br>
117    Anh Khoa
118  </footer>
119 </body>
120 </html>

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